Nickie Heimbuch

EDUCATION

University of Pittsburgh, Dietrich School of Arts and Sciences **April 2023**

Bachelor of Science in Ecology and Evolutionary Biology, Chemistry Minor, GIS Certificate

Honors: *summa cum laude*

Relevant Coursework, Licenses and Certifications:

Certificate in GIS: As part of the GIS Certificate I completed an independent study using ArcGIS Pro and LANDSAT 9 data to produce NDVI and recovery analysis maps of wildfires across Montana. I was also trained in remote sensing analysis using ENVI Classic.

Course Work: Ecology Lab, Geology Lab, Organic Chemistry Lab, Environmental Geochemistry, Biochemistry, Plant Biology, Behavioral Ecology, Biostatistics, Research Writing, Remote Sensing, Public Speaking, Photography

PUBLICATIONS

Heimbuch, N.G., McGranahan, D.A., Wonkka, C.L., Vermeire, L.T., Branson, D.H., 2023. Grasshopper abundance and offtake increase after prescribed fire in semi-arid grassland. Int. J. Wildland Fire 32, 1828–1833. https://doi.org/10.1071/WF23031

RESEARCH AND WORK EXPERIENCE

Data Analyst **Pittsburgh, PA**

Collaboration with Dr. Carissa Wonkka, University of Florida **April 2026-Present**

I am a self directed data analyst processing ecological data from Dr. Wonkka's fieldwork in eastern Montana, with the goal of producing a manuscript for publication.

- Self-trained in NMDS ordination statistical analysis, data imaging, and coding in R using the Vegan package.

Lab Volunteer **Pittsburgh, PA**

Dr. Richards-Zawacki Lab, University of Pittsburgh **April 2026-Present**

To further my education and experience with amphibian research models I have begun volunteering my time with the Richards-Zawacki Lab.

- I am learning research protocols for amphibian field research and microbiome analysis through discussion with the PhD candidates and weekly lab meetings.

Student Conservation Association Fellow **Pittsburgh, PA**

Community Greening Department, Western Pennsylvania Conservancy **September-December 2025**

I am a Tree Tender certified forestry technician focusing on urban tree planting and education with the public.

- Mapped tree planting sites in GIS and assessed locations for suitability on a per species basis
- Lead dozens of tree planting demonstrations for community volunteers and other professionals.

Biological Science Aide **Yasuní, Ecuador**

Tiputini Biodiversity Field Station, Universidad San Francisco de Quito **March 2025**

I was part of a month-long field assessment looking for potential long term botanical research sites within the Yasuní Biosphere Reserve.

- Evaluated sites based on small and large stem floral composition and terrain characteristics such as drainage, distance to water sources, grade, and navigability.
- Helped collect recordings and professional grade images of birds for an ornithological census database.
- Shadowed primatologists in the area to observe behavior of spider monkey troops and data collection techniques.

Baker **Pittsburgh, PA**

La Gourmandine French Bakery **January 2024-June 2025**

I was a baker and front of house waiter after graduating from the University of Pittsburgh and concluding my field season.

- Primarily responsible for baking almond croissants.
- Trained in kitchen safety and cleanliness protocols.

Biological Science Research Intern **Miles City, MT**
Fort Keogh Range Research Lab, USDA **May 2021- February 2022, June-September 2023**

During two separate internships, I contributed to research on mixed grassland ecology, focusing on pyric herbivory patterns, plant species identification, and soil health.

- Designed and conducted a grasshopper pyric herbivory study from hypothesis development to manuscript publication as first author in the *International Journal of Wildland Fire*.
- Presented my research to over 60 stakeholders and colleagues at the conclusion of my internships.
- Applied tidyverse R programming for statistical analysis of crude protein, grasshopper offtake, and grasshopper abundance data which I collected singlehandedly in the field.
- Collected, dried, identified, and processed grass and forb species from over 18 plots, using Avenza to ensure precise location tracking and sample location marking.
- Used a Li-Cor LI-6800 Portable Photosynthesis System to collect photosynthetic and environmental data for other scientist's research projects.
- Learned to identify dozens of key species including purple threeawn, western wheatgrass, Kentucky bluegrass, Japanese cheatgrass, silver sagebrush, Wyoming big sage, threadleaf sedge, etc.
- Built and deployed over 20 GPS collars using Arduino processors and Pelican Cases to monitor cattle movement in burned vs. unburned pasture, identifying pyric herbivory patterns and links to plant census data and soil type.
- Measured, harvested, dried, and weighed hundreds of silver sage to develop a plant size to biomass formula.
- Deployed pollinator traps and collected and stored specimens for shipment and identification in another research office.
- Collected and preserved topsoil cores for microbial analysis and soil health diagnosis.

REU Intern **Lake Mývatn, Iceland**
Ives Research Lab, University of Wisconsin Madison **June 2022- August 2022**

I collaborated on ecological projects focused on the phenology of terrestrial plants and the feeding behavior of midges, gaining hands-on experience with both field and lab work.

- Led an independent project on the dietary preferences of the midge *Tanypodinae islandicus* across four instars, designing and conducting a lab experiment, including a new dissection protocol for gut content analysis which was adopted for future research in the lab.
- Engaged with the local community in an annual open house, presenting our various limnology research projects to all ages.
- Assisted a PhD student on a terrestrial plant phenology project simulating midge population booms by fertilizing plots with midge carcasses and monitoring for differentiation in count and timing of buds, flowers, and fruiting bodies.
- Trained in identification of adult midge species through microscopic analysis of male genitalia, and preserved specimens in ethanol for long-term storage. Also learned to identify larvae instars through key morphological differences in the head capsule.
- Learned to identify benthic and pelagic zooplankton and rotifers and benthic and pelagic algae and diatom species.
- Constructed over a dozen emergence traps for in-lake deployment and gained experience with various equipment including SONDE dataloggers, sediment corers, and fluorometers, DO probes, pH meters, chlorophyll water probes, benthic zooplankton samplers, etc.

Project Leader **Pittsburgh, PA**
Stephenson Lab of Disease Ecology and Parasitology, University of Pittsburgh **January 2020-April 2020**

I led a research project exploring the relationship between parasite load and shoaling behavior in guppies, using advanced video and AI-based analysis tools.

- Engineered a staging arena equipped with FLIR HD cameras for video capture, analyzed using the AI-based software IDtracker. Videos were analyzed to quantify changes in shoaling time with parasite load.
- Managed guppy care and husbandry, ensuring proper environmental conditions such as water quality, temperature, and light cycles.
- Anesthetized guppies to count visible parasites, then resuscitated them for behavioral experiments.

Field Technician **La Milpa, Belize**

Dr. Nicholas Brokaw Botanical Census Fieldwork, Programme for Belize, Central America **February 2019**

As part of a tropical botanical census project, I collected data on flora in remote forest transects under challenging environmental conditions.

- Tagged flora, mapped large tree locations, and quantified liana and epiphyte loads along forest transects.
- Gained proficiency in using GPS equipment and compass navigation to access remote research sites and return safely.
- Trained in venomous snake identification and removal protocol ensure team safety.
- Hiked over 8 hours per day through undisturbed terrain to reach transects while maintaining vigilant situational awareness.

LEADERSHIP AND TRAINING

President **Pittsburgh, PA**

Epsilon Eta Delta: Environmental Honors Fraternity, University of Pittsburgh **August 2020–April 2023**

- Planned service projects to rejuvenate over 15 tree plots on local public streets.
- Created weekly lessons in native plant and animal identification as well as proper backpacking procedures.
- Community leader for organizing environmental activism and involvement on Pitt's campus.
- Collaborated with other local organizations to increase student volunteer participation and community engagement.

Fire Science Research Cadre **Dunn Ranch Prairie, MO**

Tallgrass Prairie and Oak Savanna Fire Science Consortium, The Nature Conservancy **April 2023**

- Wildland Fire Fighter Type 2 trained by TNC burn boss Ryan Gauger, certified in ICS-100, L-180, S-130, and S-190 training modules.
- Ignited and held fire lines during two Rx burn plans that the cadre produced and presented to the burn boss, measuring 10 and 100 acre burn areas.
- Maintained communication lines and spun weather reports during the prescription plan.

Undergraduate Teaching Assistant **Pittsburgh, PA**

Biostatistics, University of Pittsburgh **September 2022-December 2022**

- Troubleshot with students' R coding during class time and helped with lesson planning each week.
- Held weekly office hours for one-on-one instruction in R studio, including some extracurricular lessons in tidyverse for excited students.